

Question 1:

Given the following matrices

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & -1 & 0 \\ 3 & -1 & 3 \\ 2 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -1 & 6 & -1 \\ 3 & -1 & 0 & 2 \\ 1 & 1 & 2 & 4 \end{bmatrix},$$

compute the products AB , BA and $A^T A$.

$$AB = \begin{bmatrix} 12 & 1 & 14 & 19 \\ 3 & -2 & 18 & -5 \\ 6 & 1 & 24 & 7 \\ 7 & -3 & 12 & 0 \end{bmatrix}$$

$$BA = \begin{bmatrix} 15 & -2 & 26 \\ 4 & 9 & 12 \\ 18 & 3 & 10 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & -1 & -1 & 1 \\ 4 & 0 & 3 & 0 \end{bmatrix} \quad \therefore A^T A = \begin{bmatrix} 23 & -2 & 13 \\ -2 & 7 & 5 \\ 13 & 5 & 25 \end{bmatrix}$$

Question 2:

Suppose that

$$A = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 4 & 1 & 2 & 4 \\ -5 & -2 & -6 & -4 \\ 2 & -1 & -6 & x \end{bmatrix}.$$

Find a value for x such that A is singular.

~~$$|A| = -6x - 16 + 40 + 48 - (-8) - 24 - 96 - (-10x)$$~~

~~$$= -6x + 72 - 112 + 10x$$~~

~~$$= 4x - 40$$~~

~~$$\therefore |A| = 0$$~~

~~$$\therefore x = 10$$~~

$$A = LU$$

$$= \begin{bmatrix} 4 & & & \\ -5 & & & \\ 2 & & & \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & -3 & -6 & 0 \\ 0 & 3 & 4 & 1 \\ 0 & -3 & -10 & x-2 \end{bmatrix} = \begin{bmatrix} 4 & & & \\ -5 & -1 & & \\ 2 & 1 & & \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & -3 & -6 & 0 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & -4 & x-2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & & & \\ 4 & 1 & & \\ -5 & -1 & 1 & \\ 2 & 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & -3 & -6 & 0 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & x-4 \end{bmatrix}$$

$\therefore x = 4$ so final pivot is missing

Question 3:

Suppose that

$$A = \begin{bmatrix} 2 & 3 & 1 & 2 \\ 0 & 3 & 1 & 2 \\ 2 & 0 & 2 & 3 \\ 4 & 3 & -5 & -5 \end{bmatrix}.$$

- i) Compute the LU factorization of A .
- ii) Find the solution to $Ax = b$, where

$$b = \begin{bmatrix} 0 \\ -2 \\ 0 \\ 4 \end{bmatrix}.$$

i) $A = LU$

$$\begin{aligned} &= \begin{bmatrix} & & & \\ 0 & & & \\ 1 & & & \\ 2 & & & \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 & 2 \\ 0 & 3 & 1 & 2 \\ 0 & -3 & 1 & 1 \\ 0 & -3 & -7 & -4 \end{bmatrix} = \begin{bmatrix} & & & \\ 0 & & & \\ 1 & -1 & & \\ 2 & -1 & & \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 & 2 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & -6 & -7 \end{bmatrix} \\ &= \begin{bmatrix} & & & \\ 0 & & & \\ 1 & -1 & & \\ 2 & -1 & -3 & \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 & 2 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & -1 & 1 & 0 \\ 2 & -1 & -3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 & 2 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 2 \end{bmatrix} \end{aligned}$$

Question 3:

Suppose that

$$A = \begin{bmatrix} 2 & 3 & 1 & 2 \\ 0 & 3 & 1 & 2 \\ 2 & 0 & 2 & 3 \\ 4 & 3 & -5 & -5 \end{bmatrix}.$$

- i) Compute the LU factorization of A .
- ii) Find the solution to $Ax = b$, where

$$b = \begin{bmatrix} 0 \\ -2 \\ 0 \\ 4 \end{bmatrix}.$$

ii) $Ax = b$

$\therefore LUx = b \quad \therefore Lc = b$

$\therefore \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & -1 & 1 & 0 \\ 2 & -1 & -3 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 0 \\ 4 \end{bmatrix} \quad \therefore \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ -2 \\ -4 \end{bmatrix}$

$\therefore \begin{bmatrix} 2 & 3 & 1 & 2 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ -2 \\ -4 \end{bmatrix} \quad \therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ -2 \end{bmatrix}$

Question 4:

A *permutation matrix* P is a square matrix with a single 1 in every row and column, with zeros elsewhere. Show that a permutation matrix has the property $P^T = P^{-1}$.

$$\cancel{PP^T = I}$$

$$\cancel{PP^{-1} = I}$$

$$\therefore \cancel{P^T = P^{-1}}$$

Let n th column = P_n

$$\therefore P = [P_1, P_2, P_3, \dots, P_n]$$

$$\therefore P^T = \begin{bmatrix} P_1^T \\ P_2^T \\ P_3^T \\ \dots \\ P_n^T \end{bmatrix}$$

$$\therefore P^T P = \begin{bmatrix} P_1^T P_1 & P_1^T P_2 & P_1^T P_3 & \dots & P_1^T P_n \\ P_2^T P_1 & P_2^T P_2 & P_2^T P_3 & \dots & P_2^T P_n \\ P_3^T P_1 & P_3^T P_2 & P_3^T P_3 & \dots & P_3^T P_n \\ \dots & \dots & \dots & \dots & \dots \\ P_n^T P_1 & P_n^T P_2 & P_n^T P_3 & \dots & P_n^T P_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix} = I$$

$$\therefore P^{-1} P = I \quad \therefore P^T = P^{-1}$$

Question 5:

The *trace* of a matrix A is the sum of the elements on its diagonal, and is denoted $\text{tr}(A)$. If $A \in \mathbb{R}^{m \times n}$ with entries a_{ij} , show that

$$\text{tr}(A^T A) = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2,$$

i.e. show that $\text{tr}(A^T A)$ is the sum of the squares of the elements of A .

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

$$\therefore A^T A = \begin{bmatrix} a_{11}^2 + a_{21}^2 + a_{31}^2 & B & C \\ D & a_{12}^2 + a_{22}^2 + a_{32}^2 & F \\ G & H & a_{13}^2 + a_{23}^2 + a_{33}^2 \end{bmatrix}$$

$$\therefore \text{tr}(A^T A) = a_{11}^2 + a_{21}^2 + a_{31}^2 + a_{12}^2 + a_{22}^2 + a_{32}^2 + a_{13}^2 + a_{23}^2 + a_{33}^2$$

$$= \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2$$

Question 6:

When multiplying matrices and vectors, an important practical consideration is the *order* in which the operations are carried out. Suppose that

$$A = \begin{bmatrix} 1 & -2 \\ 2 & 0 \\ 3 & 1 \\ -2 & 4 \\ 0 & -1 \end{bmatrix},$$

and we are interested in the matrix product $B = A^T A A^T A$.

i) Compute B using

$$B = A^T (A A^T) A.$$

ii) Compute B using

$$B = (A^T A)(A^T A).$$

Your answer should be the same using either method. Which is easier and why?

$$i) \quad A^T = \begin{bmatrix} 1 & 2 & 3 & -2 & 0 \\ -2 & 0 & 1 & 4 & -1 \end{bmatrix}$$

$$\therefore A A^T = \begin{bmatrix} 1 & -2 \\ 2 & 0 \\ 3 & 1 \\ -2 & 4 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & -2 & 0 \\ -2 & 0 & 1 & 4 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 2 & 1 & -10 & 2 \\ 2 & 4 & 6 & -4 & 0 \\ 1 & 6 & 10 & -2 & -1 \\ -10 & -4 & -2 & 20 & -4 \\ 2 & 0 & -1 & -4 & 1 \end{bmatrix}$$

$$\therefore A^T (A A^T) = \begin{bmatrix} 1 & 2 & 3 & -2 & 0 \\ -2 & 0 & 1 & 4 & -1 \end{bmatrix} \begin{bmatrix} 5 & 2 & 1 & -10 & 2 \\ 2 & 4 & 6 & -4 & 0 \\ 1 & 6 & 10 & -2 & -1 \\ -10 & -4 & -2 & 20 & -4 \\ 2 & 0 & -1 & -4 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 32 & 36 & 47 & -64 & 7 \\ -51 & -14 & 1 & 102 & -22 \end{bmatrix}$$

Question 6:

When multiplying matrices and vectors, an important practical consideration is the *order* in which the operations are carried out. Suppose that

$$A = \begin{bmatrix} 1 & -2 \\ 2 & 0 \\ 3 & 1 \\ -2 & 4 \\ 0 & -1 \end{bmatrix},$$

and we are interested in the matrix product $B = A^T A A^T A$.

i) Compute B using

$$B = A^T (A A^T) A.$$

ii) Compute B using

$$B = (A^T A)(A^T A).$$

Your answer should be the same using either method. Which is easier and why?

$$\therefore A^T (A A^T) A = \begin{bmatrix} 32 & 36 & 47 & -64 & 7 \\ -51 & -14 & 1 & 102 & -22 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & 0 \\ 3 & 1 \\ -2 & 4 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 373 & -280 \\ -280 & 533 \end{bmatrix}$$

$$\text{ii) } A^T A = \begin{bmatrix} 1 & 2 & 3 & -2 & 0 \\ -2 & 0 & 1 & 4 & -1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & 0 \\ 3 & 1 \\ -2 & 4 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 18 & -7 \\ -7 & 22 \end{bmatrix}$$

$$\therefore (A^T A)(A^T A) = \begin{bmatrix} 18 & -7 \\ -7 & 22 \end{bmatrix} \begin{bmatrix} 18 & -7 \\ -7 & 22 \end{bmatrix} = \begin{bmatrix} 373 & -280 \\ -280 & 533 \end{bmatrix}$$

Question 7:

Suppose that P , Q and R are matrices of appropriate dimension. Show that

i) $(I + P)^{-1} = I - (I + P)^{-1}P$

ii) $(I + QR)^{-1}Q = Q(I + RQ)^{-1}$,

where all of the necessary inverses are assumed to exist.

i) $(I + P)(I + P)^{-1} = I$

$$I(I + P)^{-1} + P(I + P)^{-1} = I$$

$$\therefore (I + P)^{-1} = I - (I + P)^{-1}P$$

ii) $(I + QR)^{-1}(I + QR) = I$

$$(I + RQ)(I + RQ)^{-1} = I$$

$$\therefore (I + QR)^{-1} = I - (I + QR)^{-1}QR$$

$$(I + RQ)^{-1} = I - RQ(I + RQ)^{-1}$$

$$\therefore (I + QR)^{-1}QR = RQ(I + RQ)^{-1}$$

$$\therefore (I + QR)^{-1}Q = Q(I + RQ)^{-1}$$